

Compositional Feature Interpretability for Diffusion Language Models

Applying NIMO on Sparse Autoencoder Features
to Decompose Linear and Nonlinear Feature Interactions
along the Denoising Trajectory

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Abstract

Diffusion language models (dLLMs) predict masked tokens from full bidirectional context via iterative denoising—a paradigm fundamentally different from autoregressive (AR) language models. While sparse autoencoders (SAEs) have recently been shown to extract interpretable features from dLLM activations, how these features *compose* to drive predictions remains an open question. We propose applying NIMO—a Nonlinear Interpretable Model that decomposes predictions into global linear coefficients (β) and local nonlinear corrections (g_u)—to SAE features extracted from dLLMs.

The central argument of this proposal is that dLLMs are not merely an application domain for NIMO, but the *ideal* domain for it. First, NIMO’s “all other features” interaction structure ($\mathbf{x}_{-j}$) is structurally isomorphic to the bidirectional masked prediction mechanism of dLLMs—a property that AR models do *not* share. Second, the stochastic masking process provides controlled variation in the feature context across timesteps, enabling a stronger causal identification of global versus local effects than any standard tabular dataset. We hypothesize a *phase transition* from a β -dominated (linear, sparse context) regime at early denoising timesteps to a g_u -dominated (nonlinear, rich context) regime at late timesteps, a phenomenon unique to dLLMs and unobservable in AR models.

Introduction

Diffusion language models (dLLMs) have emerged as a compelling alternative to autoregressive language models for text generation [austin2021structured](#), [sahoo2024simple](#), [nie2025llada](#). Unlike AR models, which generate tokens sequentially from left to right using only causal context, dLLMs operate bidirectionally: they predict *all* masked tokens simultaneously using the full available context at each denoising step. This architectural difference has profound consequences for how information is represented and composed—consequences that interpretability research is only beginning to uncover.

The interpretability of dLLMs is in its early stages. DLM-Scope [dlmscope2026](#) recently demonstrated that sparse autoencoders (SAEs), powerful tools for extracting monosemantic features from AR language models [cunningham2023sparse](#), [bricken2023monosemanticity](#), can be applied to dLLMs as well. SAE features in dLLMs can steer generation, help optimize decoding order, and remain stable across post-training. Strikingly, DLM-Scope found that

inserting SAEs *reduces* cross-entropy loss in early dLLM layers—a phenomenon absent in AR models, suggesting that early dLLM representations are more linearly structured than their AR counterparts. This observation is intriguing but currently lacks a mechanistic explanation.

However, DLM-Scope studies SAE features largely in isolation. It does not address how features *compose*: which features have globally consistent linear effects, which only contribute in the presence of specific context features, and how this compositional structure evolves as the denoising process reveals more context. These are precisely the questions that NIMO nimo2026—a Nonlinear Interpretable Model—was built to answer.

This proposal argues that NIMO and dLLMs are not merely compatible but *structurally matched* at a deep architectural level. The masked prediction mechanism of dLLMs instantiates the same bidirectional “all other features” dependency that is central to NIMO’s design. Furthermore, the iterative masking process provides the controlled variation in feature context that NIMO needs to cleanly separate global effects from local interactions. This degree of controlled variation is unavailable in standard tabular settings and fundamentally inaccessible in AR models. We therefore propose that dLLMs, equipped with SAE-extracted features, constitute the natural and strongest setting for deploying NIMO as an interpretability tool.

Background

NIMO: A Nonlinear Interpretable Model

NIMO nimo2026 combines the global interpretability of linear regression with the expressiveness of neural networks. Given input features $\mathbf{x} \in \mathbb{R}^d$, the model predicts:

$$f(\mathbf{x}) = \beta_0 + \sum_{j=1}^d x_j \beta_j (1 + g_{\mathbf{u}}(\mathbf{x}_{-j}, E_j)), \quad (1)$$

where $\mathbf{x}_{-j}$ is the vector $\mathbf{x}$ with its j -th component set to zero, E_j is a fixed positional encoding identifying feature j , and $g_{\mathbf{u}}$ is a shared neural network. The fundamental constraint $g_{\mathbf{u}}(\mathbf{0}, E_j) = 0$ ensures that when no contextual signal is present, the model reduces to a linear model with coefficient β_j .

The global linear coefficients β are interpretable as marginal effects at the mean ($\text{MEM}_j = \beta_j$), providing guaranteed population-level feature importance. The correction $g_{\mathbf{u}}(\mathbf{x}_{-j})$ captures how feature j ’s contribution is modulated by the context of all other features—instance-level, nonlinear, and context-dependent. Training proceeds via parameter elimination: β admits a closed-form solution given the neural network parameters $\mathbf{u}$, and the joint problem reduces to optimizing over $\mathbf{u}$ alone. Adaptive ridge regression (equivalent to Lasso at the optimum) enforces sparsity in β , identifying a compact set of globally relevant features.

Diffusion Language Models

Diffusion language models learn to reverse a masking process. At timestep $t \in [0, 1]$, each token in a sequence is independently masked with probability t , yielding a partially observed sequence $\mathbf{x}^{(t)}$. The model is trained to predict all masked tokens simultaneously, conditioned on all unmasked tokens:

$$P(x_j \mid \mathbf{x}_{-j}^{(t)}), \quad \forall j \in \mathcal{M}^{(t)}, \quad (2)$$

where $\mathcal{M}^{(t)}$ is the set of masked positions. This prediction is inherently *bidirectional*: every unmasked token contributes to predicting every masked token. At $t = 1$ (fully masked) no context is available; at $t = 0$ (fully unmasked) the complete sequence is revealed. Generation proceeds by iteratively unmasking tokens from $t = 1$ to $t = 0$.

Sparse Autoencoders and DLM-Scope

Sparse autoencoders decompose neural network activations into sparse, interpretable feature dictionaries cunningham2023sparse,bricken2023monosemanticity. Given a hidden state $\mathbf{h} \in \mathbb{R}^{d_{\text{model}}}$, an SAE learns:

$$\mathbf{f} = \text{TopK}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{f} + \mathbf{b}_{\text{dec}}, \quad (3)$$

where $\mathbf{f} \in \mathbb{R}^{d_{\text{SAE}}}$ has at most $K \ll d_{\text{SAE}}$ nonzero entries. Each nonzero entry corresponds to the activation of a monosemantic, interpretable feature. DLM-Scope dlmscope2026 demonstrates that SAEs are viable for dLLMs, but does not study *feature composition*—how the sparse set of active SAE features jointly determine model predictions. This is the gap we address.

The Structural Fit Between NIMO and dLLMs

Architectural Isomorphism

The central observation motivating this proposal is a precise structural correspondence between NIMO’s feature interaction architecture and the dLLM’s masked prediction mechanism.

NIMO computes feature j ’s contribution to the prediction as:

$$\underbrace{\beta_j}_{\text{global linear}} \cdot x_j \cdot \left(1 + \underbrace{g_{\mathbf{u}}(\mathbf{x}_{-j}, E_j)}_{\text{context modulation from all others}} \right). \quad (4)$$

Feature j ’s effect is a baseline linear term modulated by a nonlinear function of *all other features* $\mathbf{x}_{-j}$. When there is no context ($\mathbf{x}_{-j} = \mathbf{0}$), the constraint $g_{\mathbf{u}}(\mathbf{0}, E_j) = 0$ reduces the prediction to the linear baseline $\beta_j x_j$.

A dLLM predicts masked token j as (Equation 2):

$$P(x_j \mid \mathbf{x}_{-j}^{(t)}), \quad (5)$$

a function of *all other token positions* $\mathbf{x}_{-j}^{(t)}$. When all other positions are also masked ($\mathbf{x}_{-j}^{(t)} \approx \mathbf{0}$), the model falls back to a context-free prior.

The “all other features” structure ($\mathbf{x}_{-j}$) is identical in both cases. The constraint behavior—linear baseline when context is absent—is also the same.

Key Observation. This structural isomorphism does *not* hold for autoregressive language models. An AR model predicts $P(x_j \mid x_1, \dots, x_{j-1})$ —a function of only *left* context, not all other tokens. NIMO’s $\mathbf{x}_{-j}$ is a bidirectional object that mirrors dLLMs but is incompatible with the causal attention of AR models.

Masking as Controlled Variation

NIMO separates β_j (global effect) from $g_{\mathbf{u}}(\mathbf{x}_{-j})$ (context-dependent effect) by exploiting variation in $\mathbf{x}_{-j}$ across the dataset. In standard tabular settings, this variation arises from natural co-variation in the data and can confound the estimates. The dLLM masking process provides this variation in a fundamentally more controlled way.

Concretely: fix a sequence $\mathbf{x}$ and a target masked position j . Across different timesteps $t \in \{t_1, t_2, \dots, t_T\}$, different subsets of the other positions are masked, yielding different

realizations of $\mathbf{x}_{-j}^{(t)}$ —all corresponding to the *same underlying text*. This is a built-in controlled ablation study: the same semantic content, with systematically varying subsets of context revealed.

Key Observation. The dLLM training process generates exactly the variation in $\mathbf{x}_{-j}$ that NIMO requires for identification—and this variation is *controlled* (same sequence, different masks), not confounded by natural data variation. NIMO’s β_j estimated from dLLM masking data has a stronger causal interpretation than in any standard tabular setting.

The Phase Transition Hypothesis

Combining the architectural isomorphism with the masking-induced variation yields the central empirical prediction of this proposal.

Phase Transition Hypothesis. As the diffusion timestep t decreases from 1 (fully masked) to 0 (fully unmasked):

- **Early denoising** ($t \approx 1$): the context $\mathbf{x}_{-j}^{(t)}$ is sparse (few unmasked tokens). By NIMO’s constraint, $g_{\mathbf{u}}(\mathbf{x}_{-j}^{(t)}) \approx 0$. Predictions are dominated by the *global linear* coefficients β —the model relies on a handful of robustly active SAE features with stable linear effects.
- **Late denoising** ($t \approx 0$): the context $\mathbf{x}_{-j}^{(t)}$ is rich (most tokens revealed). The correction $g_{\mathbf{u}}(\mathbf{x}_{-j}^{(t)})$ becomes large and varied. Predictions are dominated by *nonlinear feature interactions*—the model exploits the full bidirectional context to fine-tune its prediction.

This constitutes a measurable phase transition in the *composition regime* of the model’s predictions, quantified by the linearity ratio $\text{LR}(t)$ (Section 5.4).

This hypothesis also provides a mechanistic candidate explanation for DLM-Scope’s anomalous early-layer finding: early dLLM layers operate in the linear regime ($g_{\mathbf{u}} \approx 0$), so inserting an SAE into the computational path disrupts less of the model’s information processing, reducing rather than increasing cross-entropy loss.

Research Questions

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- RQ1. Global vs. local feature effects.** Which SAE features have stable, globally consistent linear contributions to dLLM predictions (large $|\beta_j|$), and which only matter in the presence of specific context features (small $|\beta_j|$, large $g_{\mathbf{u}}$ output)?
- RQ2. Phase transition in composition regime.** Does the linearity ratio $\text{LR}(t)$ decrease monotonically as $t \rightarrow 0$? At what timestep does the transition from β -dominated to $g_{\mathbf{u}}$ -dominated prediction occur, and is this transition sharp or gradual?
- RQ3. Causal identification.** Do features ranked by $|\beta_j|$ produce stronger causal steering effects than features ranked by activation frequency (DLM-Scope’s default) or linear probe coefficients? This tests whether NIMO’s global linear importance identifies causal features more precisely.

RQ4. Feature interaction structure. Are there systematic pairs or groups of SAE features that mutually modulate each other’s contributions via g_u ? Do these interaction patterns correspond to interpretable semantic relationships?

RQ5. dLLM vs. AR-LLM. Does the phase transition (RQ2) disappear or fundamentally change when the same pipeline is applied to an AR language model? This controls for generic language model properties and isolates effects specific to bidirectional masked prediction.

Proposed Methodology

Models and Data

dLLM: LLaDA nie2025llada or MDLM sahuo2024simple, both publicly available, at the 7B–8B scale.

AR-LLM baseline: A size-matched autoregressive model with publicly available SAEs (e.g., GPT-2-XL or LLaMA-3-8B).

SAEs: Pre-trained SAEs from DLM-Scope dlmscope2026 where available; we train additional SAEs for unmatched layers using the standard Top-K training objective.

Dataset: 50,000–100,000 text sequences of length $L = 128$ tokens, drawn from Wikipedia and a curated sentence-completion corpus with known semantic categories. The controlled subset enables validation of extracted feature semantics.

Stage 1: Feature Extraction with Masking-Induced Variation

For each sequence $\mathbf{x}$ and each target masked position j :

1. Sample $T = 5$ timesteps $t \in \{0.1, 0.25, 0.5, 0.75, 1.0\}$.
2. At each t , independently mask each non-target position with probability t , obtaining $\mathbf{x}^{(t)}$.
3. Forward-pass the dLLM; extract the hidden state $\mathbf{h}^{(l,t)} \in \mathbb{R}^{d_{\text{model}}}$ at target layer l .
4. Apply the SAE to obtain sparse feature vector $\mathbf{f}^{(l,t)} \in \mathbb{R}^{d_{\text{SAE}}}$ encoding the currently available context.
5. Record the target $y^{(t)} = \log P_{\text{dLLM}}(x_j | \mathbf{x}_{-j}^{(t)})$, the model’s log-probability of the correct token given the context at timestep t .

This yields a dataset of pairs $(\mathbf{f}^{(l,t)}, y^{(t)})$ across multiple instances of varying context richness for the *same* underlying sequence—the controlled variation that enables clean identification of β vs. g_u .

Stage 2: NIMO Fitting

For each (layer l , timestep t), fit NIMO with input $\mathbf{f}^{(l,t)}$:

$$\hat{y} = \beta_0 + \sum_{j=1}^{d_{\text{SAE}}} f_j^{(l,t)} \beta_j \left(1 + g_u(\mathbf{f}_{-j}^{(l,t)}, E_j)\right). \quad (6)$$

Training uses parameter elimination and adaptive ridge regression (Lasso-equivalent), producing sparse $\beta^{(l,t)}$ and the neural network $g_u^{(l,t)}$.

Dimensionality. d_{SAE} is large (e.g., 16,384) but SAE sparsity ensures at most $K = 32$ –128 active features per instance. NIMO’s Lasso further zeros out most β_j , leaving an interpretable active set of ~ 50 –200 globally important features. The effective problem dimensionality is thus manageable.

Multi-class extension. Using the multi-class NIMO formulation, we additionally fit models predicting the full vocabulary logit distribution (using the dLLM’s output as teacher), producing $\beta_j^{(k)}$ for each token class k . This identifies which SAE features globally drive predictions for specific token classes.

Stage 3: Analysis Suite

Linearity ratio (RQ2). For each timestep t , let $G_{\mathbf{u}}^{(t)} \in \mathbb{R}^{n \times d_{\text{SAE}}}$ be the matrix of nonlinear correction terms across instances. Define:

$$\text{LR}(t) = \frac{\|\beta^{(t)}\|_1}{\|\beta^{(t)}\|_1 + \frac{1}{n} \sum_{i=1}^n \|G_{\mathbf{u},i}^{(t)}\|_1}. \quad (7)$$

Plot $\text{LR}(t)$ vs. t across layers. A monotone decrease from $t = 1$ to $t = 0$ confirms the phase transition hypothesis. The timestep t^* where $\text{LR}(t^*) = 0.5$ identifies the transition point.

β -stability analysis (RQ1). Across timesteps, track each feature’s $|\beta_j(t)|$. Cluster features into: (a) “always-on” (stable high $|\beta_j|$ across all t), (b) “early-dominant” (high $|\beta_j|$ at $t \approx 1$, near zero at $t \approx 0$), (c) “late-dominant” (the reverse). Inspect the semantic identity of each cluster via human annotation of the SAE feature dictionaries.

Feature interaction analysis (RQ4). For each active feature j , compute $\text{Var}_i[G_{\mathbf{u},ij}^{(t)}]$ across instances. High variance indicates strong context-dependence. Identify the most interacting pairs (j', j) by fitting sparse linear models $G_{\mathbf{u},ij} \approx \sum_{j'} \alpha_{j'j} f_{j'} + \varepsilon$. Visualize as a directed interaction graph: an edge $j' \rightarrow j$ means activating SAE feature j' significantly modulates feature j ’s contribution.

Causal validation (RQ3). Rank SAE features by: (a) $|\beta_j|$, (b) activation frequency, (c) linear probe coefficient. For top- k features by each ranking, perform activation-patching steering experiments. Measure generation quality (perplexity, BERTScore, semantic coherence) as a function of steering magnitude. NIMO’s β_j ranking should yield the strongest causal response per unit of intervention.

AR-LLM comparison (RQ5). Apply the identical pipeline to the AR-LLM, replacing diffusion timesteps with prefix-length variation (shorter prefixes $\approx$ sparser context). Contrast $\text{LR}(t)$ curves. The prediction is that the AR-LLM will not exhibit a clean phase transition, because its causal attention structure breaks the bidirectional $\mathbf{x}_{-j}$ symmetry that NIMO models.

Advantages over DLM-Scope

The most fundamental distinction is conceptual: DLM-Scope asks *what features exist*; this proposal asks *how they compose*. SAEs and NIMO are complementary by design—SAEs provide the interpretable feature vocabulary; NIMO provides the compositional grammar. Together they constitute a complete two-level interpretability framework for dLLMs.

Expected Contributions

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- C1. Phase transition confirmation.** The first quantitative measurement of a linear-to-nonlinear composition regime transition in dLLMs, indexed by $\text{LR}(t)$ across diffusion timesteps. This directly explains DLM-Scope’s early-layer CE anomaly with a mechanistic account.

Table 1: Comparison of DLM-Scope dlmscope2026 and the proposed framework.

Dimension	DLM-Scope	NIMO on SAE Features (Proposed)
Unit of analysis	Individual SAE features	Feature composition and interaction
Global summary	Activation frequency	$\text{MEM} = \beta_j$, guaranteed interpretable as population-level marginal effect
Local explanation	None	$g_{\mathbf{u}}(\mathbf{x}_{-j})$: per-instance nonlinear correction
Feature interaction	Not modeled	Explicit; extractable as a directed interaction graph
Timestep analysis	Observational (CE measured per timestep)	$\text{LR}(t)$ curve; $\beta(t)$ trajectory; phase transition identification
Early-layer anomaly	Observed, unexplained	Testable mechanistic hypothesis via $\text{LR}(l, t)$ heatmap
Causal feature ranking	Activation frequency; steering per feature	$ \beta_j $ as principled global-importance-based ranking
Multi-class output	Not addressed	$\beta_j^{(k)}$: feature-to-token-class affinity via multi-class NIMO

C2. A two-level compositional interpretability framework for dLLMs. β : which SAE features always matter (population-level, globally stable). $g_{\mathbf{u}}$: which SAE features only matter with the right co-activations (instance-level, contextual). No existing method provides both simultaneously with guaranteed interpretability.

C3. Improved causal feature identification. If $|\beta_j|$ outperforms activation-frequency rankings in causal steering experiments, this establishes NIMO’s global coefficient as a principled basis for mechanistic intervention in dLLMs—improving on DLM-Scope’s default approach.

C4. Feature interaction graph. A new interpretability object: a graph of SAE features connected by mutual modulation effects, revealing the hierarchical semantic structure of dLLM representations.

C5. dLLM-specific empirical characterization. By contrast with the AR-LLM baseline, we identify which compositional properties are unique to bidirectional masked prediction and which are generic language model properties—sharpening the scientific understanding of what dLLMs do differently.

Timeline

References

Phase	Duration	Tasks
Infrastructure	Weeks 1–2	Set up dLLM inference with controlled masking; load and validate DLM-Scope SAEs; implement feature extraction pipeline across (l, t) pairs; confirm SAE reconstruction quality
Core experiments	Weeks 3–4	Fit NIMO on y_{token} target across all (l, t) pairs; generate $\text{LR}(t)$ curves and $\beta(t)$ trajectories; validate NIMO fit quality against linear probe baseline
Deep analysis	Weeks 5–6	Feature interaction analysis and graph construction; causal steering validation experiments; multi-class NIMO for full vocabulary distribution targets
Comparison	Week 7	AR-LLM baseline pipeline; statistical significance testing; ablation studies on SAE choice and layer selection
Writing	Week 8	Figures, tables, main paper draft and appendix

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